Lecture 3

Probabilities on effects

Objectives. *Probabilities on all effects, additive whenever the effects sum to at most the identity, are exactly density operators. Probabilities on projections alone have the same characterisation in dimension at least three, but not for a qubit. The quadratic trace rule permits pair interference and forces an ideal three-path interference combination to vanish.*

- Prove Busch's theorem by extending an effect probability to a normal expectation rule.
- Exhibit a qubit probability assignment satisfying projection additivity but not the trace rule.
- Distinguish commuting event operations from the full projection lattice, and derive the ideal three-path identity with its assumptions.

3.1 From expectations to probabilities

Theorem 1.7 started with a positive linear expectation rule on $B(\mathcal{H})$ and obtained a density operator. Linearity already related the means of noncommuting operators. Could a rule assigning probabilities to possible readings force that linearity? The answer depends on the events to which probabilities are assigned and on which sums must be respected. Throughout, $\mathcal{H}$ is a nonzero separable complex Hilbert space, and the observable algebra under discussion is all of $B(\mathcal{H})$.

Definition 3.1 (Effect and generalised probability)**.** An *effect* is a bounded self-adjoint operator F with $0 \le F \le \mathbb{1}$. A *generalised probability* is a map $\mu : \{F : 0 \le F \le \mathbb{1}\} \to [0,1]$ such that $\mu(\mathbb{1}) = 1$ and

$$\mu\Big(\sum_k F_k\Big) = \sum_k \mu(F_k) \quad \text{whenever } F_k \ge 0, \quad \sum_k F_k \le \mathbb{1}. \tag{3.1}$$

The countable clause applies when the positive partial sums converge strongly to some $F \le \mathbb{1}$; it does not assume a general convergence theorem for arbitrary series. Each summand is then an effect, since $0 \le F_k \le F \le \mathbb{1}$. Finite sums are included.

Every projection is an effect, but an effect need not square to itself. For example, $F = \frac{1}{2}|0\rangle\langle 0|$ describes a yes outcome that occurs half the time for $|0\rangle$ and never for the orthogonal $|1\rangle$. One may realise this by an ideal test followed by independent random rejection of half its yes results. A general account of such readings comes in Lecture 9. At present an effect is the bounded operator to which a probability will be assigned, not an assertion that its only eigenvalues are zero and one.

The sum condition is stronger than additivity on orthogonal projections. Let $E_z = (\mathbb{1}+\sigma_z)/2$, $E_x = (\mathbb{1}+\sigma_x)/2$. Although they do not commute, the effects $F_1 = E_z/2$, $F_2 = E_x/2$ satisfy

$F_1 + F_2 \leq \mathbb{1}$, since $E_z, E_x \leq \mathbb{1}$. Thus our rule requires $\mu(F_1 + F_2) = \mu(F_1) + \mu(F_2)$. The remainder $\mathbb{1} - F_1 - F_2$ is also an effect. An operational model is to toss a fair coin, read σ_z on heads and σ_x on tails, and record outcome 1 for (heads, plus), outcome 2 for (tails, plus), and outcome 3 otherwise. In QM-I's language these outcomes have effects $F_1, F_2, \mathbb{1} - F_1 - F_2$. Assigning an effect the same probability whatever procedure realises it is precisely the extra consistency premise. This does not mean that the two sharp spin components have been read simultaneously. It states a consistency requirement for probabilities on these three effects, rather than simultaneous definite values of the two spin observables.

There is also a noncontextuality assumption in assigning one number $\mu(F)$ to each effect: its probability is independent of which complete family of outcomes contains it. Together with the sum rule, this is a substantive premise, not a consequence of the word "probability" alone. For a density operator the proposed assignment is $\mu(F) = \mathrm{Tr}(\rho F)$. The next theorem shows that these are all the assignments under the stated assumptions, including for a spin one half.

Self-check 3.1

1. Why is $\mathbb{1} - E_z/2 - E_x/2$ positive even though E_z and E_x do not commute?
2. Does the definition require $\mu(E + G) = \mu(E) + \mu(G)$ for arbitrary projections E, G?

3.2 Busch's theorem

Theorem 3.2 (Probabilities on effects). *For every generalised probability on the effects of a nonzero separable $\mathcal{H}$, there is a unique density operator ρ such that $\mu(F) = \mathrm{Tr}(\rho F)$ for every effect F. Conversely every density operator defines such a generalised probability.*

Proof. First obtain homogeneity on effects. Additivity gives $\mu(0) = 2\mu(0)$, hence $\mu(0) = 0$. If $0 \leq F \leq G \leq \mathbb{1}$, then $G = F + (G - F)$ is an allowed sum. Therefore $\mu(F) \leq \mu(G)$. For a positive integer n, adding n copies of F/n gives $\mu(F/n) = \mu(F)/n$. Adding $m \leq n$ of those copies gives

$$\mu\big((m/n)F\big) = (m/n)\mu(F).$$

For any real $t \in [0, 1]$, choose rational numbers $r_k \leq t \leq s_k$ in $[0, 1]$ tending to t. Monotonicity gives $r_k\mu(F) \leq \mu(tF) \leq s_k\mu(F)$, so $\mu(tF) = t\mu(F)$. This argument derives continuity in the scalar multiplier from positivity and additivity; it does not assume that μ is initially a linear functional.

Extend to every positive bounded operator. For $B \geq 0$ choose $c > 0$ with $c \geq \|B\|$ and put $f(B) = c\mu(B/c)$. This does not depend on c: if $d \geq c$, then $B/d = (c/d)(B/c)$ and scalar homogeneity gives $d\mu(B/d) = c\mu(B/c)$. Two arbitrary allowed scales can be compared through their larger one. In particular $f(F) = \mu(F)$ for an effect, by taking $c = 1$.

For $B, C \geq 0$, choose $c \geq \|B\| + \|C\|$, with $c > 0$. The effects $B/c, C/c$ have sum at most $\mathbb{1}$, so

$$f(B + C) = c\mu((B + C)/c) = f(B) + f(C).$$

Also $f(tB) = tf(B)$ for $t \geq 0$: for $t > 0$, use tc as the scale for tB when c is a scale for B; the zero case is immediate.

Extend to self-adjoint operators without ambiguity. Every bounded self-adjoint A can be written $B - C$ with $B, C \geq 0$, for example $B = A + \|A\|\,\mathbb{1}$, $C = \|A\|\,\mathbb{1}$. Define $g(A) = f(B) - f(C)$. If also $A = B' - C'$, then $B + C' = B' + C$. Positive additivity gives $f(B) + f(C') = f(B') + f(C)$, so the value is independent of the decomposition. Adding two differences and using positive

additivity proves real additivity of g; positive scalar homogeneity and reversal of a difference prove real homogeneity, including negative scalars.

Extend to all bounded operators. Write uniquely $A = C + \mathrm{i}D$ with C, D self-adjoint, and set $\omega(A) = g(C) + \mathrm{i}g(D)$. Real linearity is already proved. Multiplication by i sends $C + \mathrm{i}D$ to $-D + \mathrm{i}C$, so $\omega(\mathrm{i}A) = \mathrm{i}\omega(A)$; thus the extension is complex linear. On positives it is f, so it is positive and $\omega(\mathbb{1}) = 1$. On effects it is precisely μ.

Normality is where countable additivity enters. For orthogonal projections (E_k) the strong sum $E = \sum_k E_k$ is a projection at most $\mathbb{1}$, as established in § 1.3. The defining sum rule gives $\omega(E) = \mu(E) = \sum_k \mu(E_k) = \sum_k \omega(E_k)$. Thus ω is normal. By Theorem 1.7 it equals $\mathrm{Tr}(\rho\,\cdot)$ for a unique density operator. The extension was forced by μ, since positives are positive multiples of effects and every bounded operator is a complex combination of positives. This also proves uniqueness for the given effect probabilities.

Conversely, verify the whole sum rule. Write $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ and let $F = \sum_k F_k \leq \mathbb{1}$ be an allowed strong sum of effects. Then

$$\mathrm{Tr}(\rho F) = \sum_j p_j \left\langle \varphi_j, F\varphi_j \right\rangle = \sum_j \sum_k p_j \left\langle \varphi_j, F_k\varphi_j \right\rangle = \sum_k \mathrm{Tr}(\rho F_k).$$

Strong convergence gives the middle equality for each vector; every summand is nonnegative, so the two scalar sums can be interchanged. Positivity, the bound by one and normalisation follow from L1. ∎

The extension steps required only finite additivity. In finite dimension this would already suffice, because every positive linear normalised functional is normal there. In infinite dimension the countable clause is essential for the conclusion to be a density operator: L1 cited the existence of non-normal algebraic states. Restricting such a functional to effects would preserve finite additivity, but not all the countable sums required here. The proof has therefore named exactly the additional assumption that makes its infinite-dimensional conclusion possible.

For a normal state and any ONB (e_k), the partial level events $E_N = \sum_{k\leq N} \left|e_k\right\rangle\!\left\langle e_k\right|$ must satisfy $\mu(E_N) \to 1$. There is no missing probability beyond all the levels. Finite additivity alone does not pass from E_N to its strong limit: $\|\mathbb{1} - E_N\| = 1$ in infinite dimension. Positive normalised functionals that vanish on every finite-rank operator do exist (· *cited*; Hall, Ex. 19.5, p. 439). Their projection assignments give $\mu(E_N) = 0$ for every N but $\mu(\mathbb{1}) = 1$. They are excluded here by the countable clause, not by an additional choice of basis or a finite-dimensional calculation.

Example 3.3 (Using the effect sum rule)**.** Suppose a qubit assignment gives $\mu(E_z) = 3/4$ and $\mu(E_x) = \mu(E_y) = 1/2$, where $E_a = (\mathbb{1} + \sigma_a)/2$. Homogeneity forces $\mu(\mathbb{1}/2) = 1/2$. Consider

$$F = \tfrac{1}{2}E_z + \tfrac{1}{4}E_x + \tfrac{1}{4}(\mathbb{1}/2) = \tfrac{1}{2}\mathbb{1} + \tfrac{1}{4}\sigma_z + \tfrac{1}{8}\sigma_x.$$

It is an effect because it is a convex combination of effects: both F and $\mathbb{1} - F$ are positive. Additivity applies to the three displayed positive summands even though the spin projections do not commute. It gives $\mu(F) = \tfrac{1}{2}(3/4) + \tfrac{1}{4}(1/2) + \tfrac{1}{4}(1/2) = 5/8$. The representing state has Bloch vector $(0, 0, 1/2)$ and density matrix $\mathrm{diag}(3/4, 1/4)$; multiplying matrices gives the same $\mathrm{Tr}(\rho F) = 5/8$.

By contrast, $B = 3F$ is not an effect. The positive extension gives $f(B) = 3\mu(F) = 15/8$, which is a possible expectation but cannot be an outcome probability. The construction distinguishes the units and range of an observable's mean from the dimensionless probability of an effect. It does not bound every positive observable's mean by one.

Self-check 3.2

1. Why can two choices of the scale c not change $f(B)$?
2. Where is countable, rather than finite, additivity first needed?
3. In the self-adjoint extension, why is choosing a different positive decomposition harmless?

3.3 Gleason, and why a qubit escapes

Suppose probabilities are assigned only to projections, with $\mu(\mathbb{1}) = 1$ and countable additivity only on orthogonal families. This does not explicitly relate noncommuting effects. In most dimensions it still has a remarkable consequence, but the proof is much deeper than the extension argument just given.

Theorem 3.4 (Gleason · *cited*). *On a separable complex Hilbert space of dimension at least three, every countably additive probability on projections is $\mu(E) = \mathrm{Tr}(\rho E)$ for a unique density operator ρ.*

This is Landsman's Theorem 4.29, p. 120. We use its statement, not an unwritten proof. Its probability rule assigns the same number to a projection in every orthogonal resolution containing it. That shared assignment across different bases supplies much of the theorem's force.

In $\mathbb{C}^2$ every nonzero proper projection has the Bloch form $E(\mathbf{n}) = (\mathbb{1} + \mathbf{n}\cdot\boldsymbol{\sigma})/2$, with $|\mathbf{n}| = 1$. Its only nonzero orthogonal partner is $\mathbb{1} - E(\mathbf{n}) = E(-\mathbf{n})$. Hence any function $h : S^2 \to [0, 1]$ satisfying $h(-\mathbf{n}) = 1 - h(\mathbf{n})$ defines a projection probability: assign $\mu(E(\mathbf{n})) = h(\mathbf{n})$, $\mu(0) = 0$, $\mu(\mathbb{1}) = 1$. Every orthogonal sum is either trivial or this complementary pair, so all required additivity conditions hold.

For a continuous example take

$$h(\mathbf{n}) = \frac{1 + n_z^3}{2}.$$

If this were $\mathrm{Tr}(\rho E(\mathbf{n}))$ for a qubit density matrix with Bloch vector $\mathbf{r}$, it would equal $(1+\mathbf{r}\cdot\mathbf{n})/2$. The x and y axes force $r_x = r_y = 0$; the positive z axis forces $r_z = 1$. But at $\mathbf{n} = (\sqrt{3}/2, 0, 1/2)$ the prescribed value is $9/16$, whereas that density matrix gives $3/4$. This contradiction proves the dimension-two failure even for a continuous assignment.

A two-valued example will be useful in Lecture 13. For a unit $\mathbf{n}$, inspect (n_z, n_y, n_x) in that order and put $h_0(\mathbf{n}) = 1$ if its first nonzero entry is positive, and zero if that entry is negative. An antipodal pair has the same first nonzero position with opposite signs, so $h_0(-\mathbf{n}) = 1 - h_0(\mathbf{n})$. This assigns only zero or one to every qubit projection while respecting each orthogonal resolution. The equator and the two remaining axis points are included by the tie-breaking order.

Neither assignment is a quantum state missed by L1. By Busch, neither extends to an effect probability with the full sum property. In particular the two-valued assignment is not a positive linear expectation rule: L2 proves that every density operator has an event with probability strictly between zero and one. Different assumptions define different classes of assignments. This is the dimension-two escape that the Kochen–Specker discussion in Lecture 13 must retain.

Self-check 3.3

1. Construct a two-valued assignment using the first nonzero entry of (n_x, n_y, n_z) instead. Why is a tie-breaker needed when $n_x = 0$?
2. Which assumption distinguishes Busch's theorem from Gleason's, and which theorem applies to a qubit?

3.4 Which events obey Boolean rules?

For projections, $E \wedge G$ denotes projection onto the intersection of their ranges; $E \vee G$ denotes projection onto their closed linear span. The complement is $\mathbb{1} - E$. The projections with these operations are called the *projection lattice*. The general existence of projections onto closed subspaces is the Hilbert projection theorem, cited in *Linear Algebra*, Lecture 13. Our computations below either construct the needed projection explicitly or take place in $\mathbb{C}^2$; they do not depend on that general cited theorem.

If $EG = GE$, then EG is self-adjoint and $(EG)^2 = EG$, with range exactly $\operatorname{ran} E \cap \operatorname{ran} G$. Likewise $E + G - EG = \mathbb{1} - (\mathbb{1} - E)(\mathbb{1} - G)$ is a projection. Its range is contained in the sum of the two ranges, and it fixes each of them, so it projects onto their sum. Thus

$$E \wedge G = EG, \qquad E \vee G = E + G - EG \quad (EG = GE).$$

For three pairwise commuting projections E, G, R, distributivity follows by multiplication:

$$E \wedge (G \vee R) = EG + ER - EGR = (E \wedge G) \vee (E \wedge R),$$

since $(EG)(ER) = EGR$. Complements, finite joins and finite meets of commuting projections still commute. These are the Boolean operations of ordinary events in one compatible family. QM-I's commutation criterion for compatibility explains their physical setting.

For instance, in a three-dimensional basis take $E = \operatorname{diag}(1, 1, 0)$ and $G = \operatorname{diag}(0, 1, 1)$. They overlap: $E \wedge G = \operatorname{diag}(0, 1, 0)$, while $E \vee G = \mathbb{1}$. In the state $\rho = \operatorname{diag}(1/2, 1/3, 1/6)$ their probabilities are $5/6$ and $1/2$, and the intersection probability is $1/3$. The ordinary inclusion–exclusion rule gives $5/6 + 1/2 - 1/3 = 1$ for the join. Simply adding the probabilities would double-count the shared event. Commutation gives a common Boolean description; it does not by itself say that two events are exclusive.

For incompatible events the same operations need not be distributive. In $\mathbb{C}^2$, let $E_1 = |0\rangle\langle 0|$, $E_2 = \big|(|0\rangle + |1\rangle)/\sqrt{2}\big\rangle\big\langle(|0\rangle + |1\rangle)/\sqrt{2}\big|$ and $E_3 = \big|(|0\rangle - |1\rangle)/\sqrt{2}\big\rangle\big\langle(|0\rangle - |1\rangle)/\sqrt{2}\big|$. Their ranges are shown in Figure 3.1.

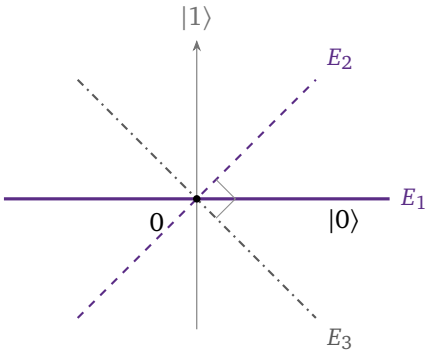


Figure 3.1: The real-coordinate section of the three complex rays. Each drawn line represents the complex span of its indicated vector: the orthogonal rays E_2, E_3 span $\mathbb{C}^2$, but each meets E_1 only at zero.

The last two rays span the plane, so $E_2 \vee E_3 = \mathbb{1}$. Each intersects the first ray only in zero. Consequently

$$E_1 \wedge (E_2 \vee E_3) = E_1 \neq 0 = (E_1 \wedge E_2) \vee (E_1 \wedge E_3).$$

These are subspace operations, not a claim that one can perform all three sharp readings without regard to their order. For instance $E_1 \vee E_2 = \mathbb{1}$, yet in the state $|0\rangle\langle 0|$ the sum of their probabilities is $1 + 1/2 = 3/2$, not one. A join of nonorthogonal rays is not generally a probability sum. The projection lattice does not provide one classical sample space for every incompatible reading at once.

Self-check 3.4

1. Why does $\mu(E_2 \vee E_3) = \mu(E_2) + \mu(E_3)$ hold in this example, but not for an arbitrary pair of rays?
2. Compute both sides of distributivity in the displayed counterexample.

3.5 Three paths, but only pair interference

Let E_A, E_B, E_C be mutually orthogonal path projections. Fix one unit preparation ψ, one subsequent unitary U and one detector effect $0 \leq D \leq \mathbb{1}$. Opening a subset S of ideal paths means applying $E_S = \sum_{j \in S} E_j$. The common experimental arrangement is shown in Figure 3.2.

The detection probability per original trial is

$$p_S = \langle UE_S\psi, DUE_S\psi\rangle = \Big\|\sum_{j\in S} \eta_j\Big\|^2, \qquad \eta_j = \sqrt{D}\,UE_j\psi.$$

The square root is S6. Blocking paths discards amplitude; we do not renormalise the transmitted state. Since $D \leq \mathbb{1}$ and E_S is a projection, $0 \leq p_S \leq 1$. There is no background in this ideal model, so $p_\varnothing = 0$.

Write $q_j = \|\eta_j\|^2$ and $b_{jk} = 2\,\mathrm{Re}\,\langle \eta_j, \eta_k\rangle$. Expanding the norm square gives $p_{jk} = q_j + q_k + b_{jk}$ and $p_{ABC} = p_A + p_B + p_C + b_{AB} + b_{AC} + b_{BC}$. Each pair contribution appears once in the triple

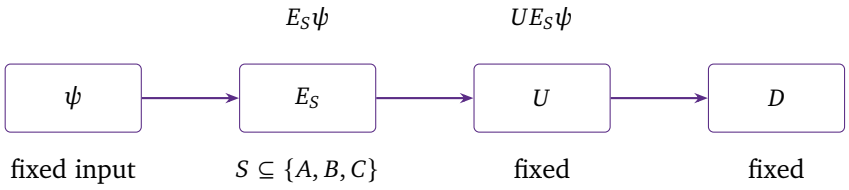


Figure 3.2: The ideal filter protocol. Only the subset S changes; the preparation, subsequent evolution and detector stay fixed. The transmitted vector $E_S\psi$ is not normalised again, and each detection count is divided by the same original-trial count to estimate its probability.

and once among the pairs; each single appears once in the triple, twice among the pairs and once as a single. Thus

$$I_3 = p_{ABC} - p_{AB} - p_{AC} - p_{BC} + p_A + p_B + p_C = 0. \tag{3.2}$$

The ordinary two-path difference $p_{AB} - p_A - p_B = b_{AB}$ can be nonzero. The third-order combination, Sorkin's I_3, adds no new term because a norm square is quadratic. Three open paths can still interfere: p_{ABC} need not equal $p_A + p_B + p_C$, but its change is entirely a sum of pair contributions.

For a numerical ideal example take $\mathcal{H} = \mathbb{C}^3$, path basis e_A, e_B, e_C, $\psi = (e_A + e_B + e_C)/\sqrt{3}$, $U = \mathbb{1}$ and $D = |\psi\rangle\langle\psi|$. Then $p_S = |S|^2/9$: each single has $1/9$, each pair $4/9$, and the triple 1. Equation (3.2) reads $1 - 3(4/9) + 3(1/9) = 0$, whereas the two-path contribution is $2/9$. If each transmitted preparation were renormalised instead, its detection probability would be $|S|/3$ in this particular example. Here the conditioned I_3' happens still to vanish; the next detector shows why it need not. Such conditional probabilities describe a different protocol; they are not the quantities in the derivation. In general renormalisation destroys the common quadratic expression and its identity.

One can see a false residual using the same uniform preparation but the detector $D = |e_A\rangle\langle e_A|$. Before conditioning, $p_S = 1/3$ if $A \in S$ and zero otherwise, and $I_3 = 0$. The probability that any path passes is $|S|/3$. Dividing by that separate transmission probability gives $p_S' = 1/|S|$ when $A \in S$ and zero otherwise. These conditional numbers instead give $I_3' = 1/3 - 1/2 - 1/2 + 1 = 1/3$. All calculations obey Born's rule; the nonzero value comes from combining different conditioned preparations. This is why the common original trial count is a physical part of the proposed test, not merely a convention for writing p_S.

The physics

Sinha and collaborators tested a three-slit arrangement in 2010 and reported a bound below 10^{-2} of the expected two-path interference scale for the third-order contribution. The experiment includes a background measurement with all slits closed. Real slits approximate the ideal filters: changing an aperture can also change propagation and boundary conditions. Interpreting a residual therefore requires control of those changes and detector backgrounds, rather than treating any nonzero raw combination as a failure of Born's rule. The paper is *Science* **329**, 418–421, doi:10.1126/science.1190545.

The assumptions making the mathematical test exact are now visible: the preparation, evolution and detector are held fixed; the filters add as orthogonal projections; and probabilities use the same original trial count. A different unitary for each slit configuration is outside this identity. This is how a theorem becomes a proposed experimental comparison:

its idealisation must be checked along with its numbers.

Self-check 3.5

1. Verify $p_S = |S|^2/9$ in the three-dimensional example.
2. Which assumption fails if a separate transmitted state is normalised for each open-path configuration?

Probabilities and states

A countably additive probability on all effects, additive on every positive family summing to at most the identity, is exactly $\mathrm{Tr}(\rho\,\cdot)$. On projections alone, Gleason gives this conclusion in dimension at least three; a qubit admits other assignments. Commuting event projections obey Boolean rules; the full collection need not. An ideal self-adjoint observable's events are its spectral projections; together with those of an incompatible observable they need not form a Boolean algebra. The quadratic trace rule forces $I_3 = 0$ for the specified ideal path-filter protocol.

Summary. Busch derives linearity from the effect sum rule, then uses countable additivity to reach L1's normal states. Gleason reaches the same trace rule from fewer events but needs dimension at least three. Neither theorem derives its premises from probability alone. The event counterexample separates incompatible contexts, and the path calculation turns the probability law into a testable ideal identity. Next we ask which changes of rays preserve their transition probabilities.

Exercises. Try Sheet 3 in the separate *Exercise sheets* booklet (English).

Self-check answers.

3.1 1. $E_z + E_x \leq 2\mathbb{1}$ by adding the two positive inequalities. 2. Only if their sum is an effect; two projection summands then must be orthogonal.

3.2 1. Compare both scales through the larger and use real homogeneity on effects. 2. In normality on orthogonal projection sums. 3. Cross-add the two positive decompositions and use additivity of f.

3.3 1. Give value one when the first nonzero entry is positive. Without a tie-breaker, both vectors of an antipodal pair on $n_x = 0$ would receive zero. 2. Busch relates all summable effects, including noncommuting ones; it also applies in dimension two.

3.4 1. E_2, E_3 are orthogonal, whereas arbitrary rays need not be. 2. The left is E_1 and the right is zero.

3.5 1. $\langle\psi, E_S\psi\rangle = |S|/3$, and square its modulus. 2. A configuration-dependent divisor replaces the common unnormalised quadratic expression.

Sources. Landsman, *Foundations of Quantum Theory*, Thm 2.44 pp. 73–74 (finite extension proof), §2.10 pp. 75–77, Thm 4.29 p. 120 (Gleason). P. Busch, *Phys. Rev. Lett.* **91**, 120403 (2003), arXiv:quant-ph/9909073. R. Sorkin, *Mod. Phys. Lett. A* **9**, 3119–3128 (1994), arXiv:gr-qc/9401003. U. Sinha et al., *Science* **329**, 418–421 (2010), arXiv:1007.4193. The infinite-dimensional normality step uses L1 directly, not a trace-duality theorem.